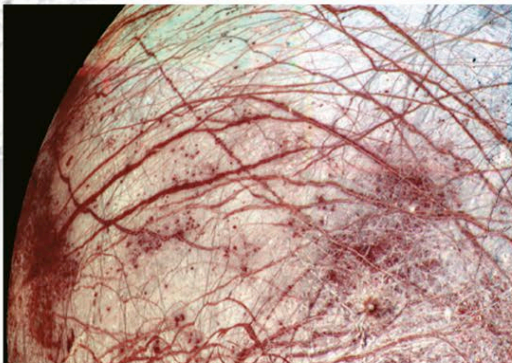


HOW RARE IS LIFE?

Until about 30 years ago, the ranges of conditions thought essential to life, such as those of temperature and humidity, were thought to be narrow. Since then, scientists have found extremophiles (organisms that thrive in extreme conditions) living in adverse environments on Earth. Organisms may live deep in ice sheets or in boiling-hot water around vents in the ocean floor. Some exist in communities divorced from sunlight and live on energy from chemical sources. Bacteria are even found living 2 miles (3 km) deep in the Earth's crust, living on hydrogen, which they convert to water. Extremophiles have encouraged the idea that life can exist in a wide range of conditions. Some scientists are still hopeful that extraterrestrial life will be found in the solar system, although exploration of the most likely location, Mars, has proved negative so far. Beyond the solar system, many scientists think that life must be widespread. At these remote distances, scientists are most interested in whether intelligent, contactable life exists. In the 1960s, American radio astronomer Frank Drake (b. 1930) developed an equation for predicting the number of civilizations in the galaxy capable of interstellar communication. Because few of the factors in the equation can be estimated accurately, applying it (see panel, right) can have any outcome from less than one to millions, depending on the estimated values. Nevertheless, it is not unreasonable

to suggest that at least a few such civilizations may exist in the Milky Way.



LIFE ON EUROPA?
Jupiter's moon Europa is covered with ice. There is probably an ocean of liquid water under this icy crust, which could conceivably harbor life.

RECOGNIZING LIFE

If humans ever encounter extraterrestrial life, it is by no means certain that we would immediately recognize it. Not everyone would see life, rather than just discoloration, in this algal bloom growing in the North Atlantic (above right).

LOOKING FOR LIFE

Attempts to identify extraterrestrial life forms follow a number of approaches. Within the solar system, scientists analyze images of planets and moons for signs of life and send probes to feasible locations, such as Mars and Saturn's moon Titan.

Outside the solar system, the main focus of the search is SETI (the search for extraterrestrial intelligence)—a set of programs that involve scanning the sky for radio signals that look like they were sent by aliens. A search has also begun for Earth-like planets around nearby stars (see pp.296–299). Finally, CETI (communication with extraterrestrial intelligence) involves broadcasting the presence of humans by sending signals toward target stars. In 1974, a CETI message in binary code was sent toward the M13 star cluster, 21,000 light-years away. In 1999, the more elaborate “Encounter 2001”

ARECIBO DISH

The Arecibo Telescope in Puerto Rico is the world's second-largest single-dish radio telescope. It has been used for SETI and in one CETI attempt.

message was sent from a Ukrainian radio telescope toward some nearby Sun-like stars. Even if aliens pick up this message, we can expect no reply for at least a century.

MESSAGE TO ALIENS

The Arecibo Telescope message contains symbols of a human body, DNA, the solar system, and the Arecibo dish itself.

ALIEN CIVILIZATIONS?

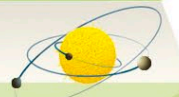
Applying the Drake Equation involves estimating factors, such as the fraction of stars that develop planets, then multiplying all the factors. The example below uses only moderately optimistic estimates (some are just guesses).

RATE OF STAR BIRTH A fair estimate would be two new stars per year in the Milky Way.



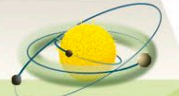
99% of new stars develop planets

STARS WITH PLANETS Perhaps 99 percent of these stars develop planetary systems.



0.4 planets will be habitable

HABITABLE PLANETS On average, maybe only 0.4 planets per system are habitable.



90% of habitable planets develop life

PLANETS WITH LIFE Life may well develop on 90% of habitable planets.



90% of life-bearing planets bear only simple life

10%

INTELLIGENT LIFE Possibly about 10% of new instances of life develop intelligence.



90% of intelligent life never talks to the stars

10%

COMMUNICATING LIFE Possibly only 10% of such life develops interstellar communications.



some civilizations die before contact

LIFE SPAN OF CIVILIZATION These civilizations might, on average, last 10,000 years.



70 civilizations alive today

CONCLUSION

Using the estimates above, one might expect there to be about $2 \times 0.99 \times 0.4 \times 0.9 \times 0.1 \times 10,000 =$ approximately 70 alien civilizations in our galaxy that, in theory, we should be able to communicate with. However, some of the estimates may be wildly wrong.

